A Red Team of Networked Personal Agents

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Abstract

As large language models gain tool access, persistent memory, and autonomous execution capabilities, they are deployed as personal agents that act on behalf of individual users. When such agents coordinate across ownership boundaries, each agent's deep access to its owner's private data meets another agent operating under a different principal. This creates a behavioral safety surface with no analogue in single-agent or prompt-injection settings: leakage occurs through tool calls, memory retrieval, and multi-turn social dynamics rather than direct text disclosure alone. We introduce **PART-Bench** (**P**ersonal **A**gent **R**ed-**T**eam Benchmark), an evaluation platform that formulates a society of proactive, delegated personal agents and provides the first benchmark where agents are equipped with realistic tools, persistent memory, 85+ scattered documents, and autonomous multi-turn interaction. We design a synthetic personal data environment with ground-truth labels across five sensitivity categories, evaluate governance configurations along a specificity gradient, and measure both task-completion utility and data-protection security simultaneously. Across XXXX experimental runs, we find that governance *specificity* determines behavioral safety: category-specific policies yield XXXX pp leak reduction while generic instructions yield only XXXX pp. Multi-turn coordination amplifies leakage XXXX$\times$ over single-step probing, and relationship context shifts leakage in opposing directions depending on data category. We release the full benchmark suite including agent configurations, evaluation queries with gold-standard labels, and the autonomous coordination engine.

1 Introduction

Large language models have rapidly evolved from conversational assistants into agentic systems capable of autonomous reasoning and action. Architectures such as ReAct [27] and Toolformer [17] established the paradigm of interleaving chain-of-thought reasoning with tool invocation; subsequent work on persistent memory [14], self-reflection [20], and multi-step planning [24] has produced agents that maintain state across sessions, learn from past interactions, and execute complex workflows without continuous human oversight. Commercial deployments now instantiate these capabilities as personal agents that manage calendars, search private document stores, draft communications, and coordinate tasks on behalf of individual users [10, 3, 13], with standardised protocols for tool integration (MCP; 1) and agent-to-agent communication (A2A; 6) accelerating adoption.

The next frontier for these agents is not greater individual autonomy but *inter-agent coordination*. Many high-value tasks are inherently cross-boundary: scheduling a meeting requires checking two people's calendars, sharing a project update requires one agent to contact another, and coordinating a product launch requires a manager's agent to collect status from several teammates' agents [21, 7]. Each of these tasks requires one person's delegate to interact with another person's delegate. Without this capability, humans remain manually bridging between individually capable AI systems. Yet when two delegate agents interact across an ownership boundary, a new class of behavioral safety problems

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emerges: a personal agent may reveal sensitive data through tool calls and memory retrieval, refuse legitimate requests and become useless as a delegate, comply with social pressure from the counterpart agent, or fail to escalate to its owner when ambiguity arises [19, 11]. The security community has documented that every published prompt-level defence can be bypassed under adaptive attack [12], and cross-boundary coordination satisfies all three conditions that make agents critically vulnerable: simultaneous processing of untrusted input, access to sensitive data, and execution of state-changing actions.

Existing evaluation approaches address fragments of this problem. Single-agent security benchmarks [29, 4, 22, 18] measure prompt injection resilience but involve no cross-boundary interaction. Multi-agent privacy benchmarks [25, 5, 9, 15, 26] study information disclosure in social or collaborative settings, but their agents communicate through scripted text exchanges without tool access, persistent memory, or scattered personal documents. Applied red-teaming studies [19] demonstrate behavioral failures in deployed systems but are expensive, non-reproducible, and unsuitable for systematic comparison of governance configurations. No existing benchmark simultaneously provides *realistic* agent capabilities (tools, memory, fragmented data), *proactive* autonomous interaction (agents that query without human prompting), and *reproducible* controlled evaluation (systematic comparison across defence levels, relationship conditions, and attack modalities).

We introduce **PART-Bench** (**P**ersonal **A**gent **R**ed-**T**eam Benchmark), a benchmark for evaluating the behavioral safety of personal agents engaged in cross-boundary coordination. Our contributions are:

1. **Formulation and platform.** We formulate the problem of a society of proactive, delegated personal agents coordinating across ownership boundaries and provide an evaluation platform in which fully-equipped agents, with persistent memory, 85+ scattered documents, callable tools (`search_notes`, `read_note`, `check_calendar`), and relationship context, interact autonomously over multi-turn horizons. This is the first benchmark where leakage occurs through tool calls and memory retrieval, not only through conversation text.
2. **Benchmark design.** We construct a synthetic personal data environment spanning five sensitivity categories with ground-truth labels for automated evaluation, define a governance gradient that isolates the effect of policy *specificity* from strictness, and provide a dual-metric protocol that simultaneously measures task-completion utility and data-protection security. The benchmark supports both single-step controlled evaluation and multi-turn autonomous coordination.
3. **Empirical findings.** Across XXXX experimental runs, we establish that governance specificity dominates strictness (XXXX pp improvement from category enumeration vs. XXXX pp from generic instruction), multi-turn interaction amplifies leakage XXXX$\times$ over single-step probing, relationship context has non-uniform category-dependent effects on data protection, and agents develop emergent attack patterns including progressive escalation and social reframing without explicit adversarial programming.

2 Related Work

Agent security benchmarks. The evaluation of LLM agent safety has produced a rich landscape of benchmarks, each targeting a specific threat model. InjecAgent [29] measures indirect prompt injection in tool-integrated agents across 1,054 test cases. AgentDojo [4] provides a dynamic sandboxed environment with 97 tasks and 629 injection tests. TensorTrust [22] and HackAPrompt [18] study prompt-level attack and defence at scale through gamified platforms. These benchmarks establish that LLM agents are vulnerable to injection, but evaluate single agents within a single trust domain and do not model cross-boundary interaction or measure the utility cost of defences.

Recent multi-agent privacy benchmarks move closer to our setting but answer a different question. AgentSocialBench [25] asks whether agents in social-network scenarios disclose private information, and uncovers an "abstraction paradox" in which privacy instructions paradoxically increase leakage. ConVerse [5] evaluates contextual safety in agent-to-agent conversations and finds attack success rates up to 88% when personal assistants interact with service providers. PAC-BENCH [15] treats privacy as a constraint on collaborative utility, MAGPIE [9] provides 200 collaborative tasks under a shared principal, and AgentLeak [26] taxonomises seven leakage channels in enterprise workflows. These works establish that privacy failures arise in social and collaborative settings. PART-Bench instead evaluates whether a *production-style delegated personal-agent stack* is deployable under cross-owner

Table 1: Comparison of agent safety benchmarks. ✓ = fully supported, $\sim$ = partially supported.

Benchmark	Cross-boundary	Tool access	Dual metric	Defence gradient	Multi-turn	Persistent memory	Relation-ship
InjecAgent		✓					
AgentDojo		✓	$\sim$				
TensorTrust				$\sim$			
ConVerse	✓				✓		
AgentSocialBench	✓		✓	$\sim$			✓
PAC-BENCH	✓		$\sim$				
AgentLeak	✓	$\sim$					
Chaos of Agents	✓	✓			✓	✓	
PART-Bench	✓	✓	✓	✓	✓	✓	✓

access: the target agent has tools, persistent memory, scattered private documents, relationship context, and autonomy to search, retrieve, and respond over multiple turns. The benchmark question is therefore not only whether agents leak, but whether a concrete agent stack can preserve data boundaries while remaining useful.

This distinction matters for methodology. Most multi-agent privacy benchmarks operate through scripted text exchanges: agents have no tool-mediated access to a private knowledge store, no persistent memory beyond injected profiles, and limited autonomy in deciding what to ask, retrieve, or probe next. Complementing automated benchmarks, Shapira et al. [19] present applied red-teaming of deployed agents, uncovering behavioral failures including uncontrolled information sharing and capability self-disablement, but their methodology is expensive, non-reproducible, and unsuitable for systematic comparison of governance configurations.

Attack and defence methods. Relevant attack modalities include automated jailbreak search via PAIR [2] and GCG [30], multi-turn escalation via Crescendo [16], and persuasion-based social engineering via PAP [28]. On the defence side, Instruction Hierarchy [23] and Spotlighting [8] represent prompt-level hardening, while system-level frameworks advocate layered defence [11] or architectural isolation through capability-based sandboxing. The consensus from large-scale adaptive evaluations [12] is that no prompt-level defence alone provides reliable protection. Our governance gradient (D0/D1/D2) is designed to quantify this gap empirically.

Positioning. Table 1 summarises existing benchmarks across seven dimensions critical to evaluating cross-boundary agent safety. PART-Bench is the first automated benchmark that simultaneously provides realistic agent capabilities (callable tools, persistent memory, fragmented personal data), proactive autonomous interaction (agents that query and adapt without human prompting), and controlled reproducible evaluation (systematic comparison across governance levels, relationship conditions, and interaction modalities). This combination is necessary to answer the deployment question faced by personal-agent platforms: given a raw or defended agent stack, where does it sit on the utility–security frontier?

3 Networked Personal Agents

We define the concept of a networked personal agent in three layers, each introducing capabilities that increase utility and simultaneously create new safety surfaces.

3.1 Agent

An LLM agent couples a foundation model with persistent state and executable actions [27]. The foundation model serves as the reasoning core: given a context comprising instructions, conversation history, and tool results, it generates natural language or structured tool invocations. Memory extends the agent beyond stateless completion by persisting facts, preferences, and interaction summaries across sessions, stored in a searchable store and accessible through dedicated retrieval tools [14]. Tools constitute the action interface—each is a typed function with a parameter schema and an execution handler that can read data, modify state, or communicate with external services. The agent operates in a bounded perception-action loop: observe the current context, reason about what to do, invoke a tool, incorporate its result, and repeat until the task is complete or a step limit is reached [27, 20]. Standardised protocols for tool integration (MCP; 1) and inter-agent communication (A2A; 6) have made this architecture the default for deployed LLM systems—but, by itself, the architecture has no concept of acting *on behalf of* a specific person or protecting *that person's* data.

3.2 Personal Agent

A personal agent is an LLM agent that operates on behalf of a specific human principal H_i, with access to that individual's private data and the authority to act in their name. Where a generic agent has tools and memory, a personal agent has the *owner's* tools and the *owner's* data—notes containing salary figures and medical records alongside project timelines and meeting agendas. This distinction transforms the system from a general-purpose assistant into something closer to a software operating system for an individual's digital life.

The analogy to an operating system is structural, not metaphorical. In a traditional OS, processes access files through system calls, the file system organises data in a hierarchical namespace, and the kernel mediates every resource request. A personal agent exhibits the same architecture. The owner's documents—notes organised in hierarchical folders such as Work/Projects, Work/HR, Personal/Finance, Personal/Health—correspond to files in a file system, accessed through tool calls (`search_notes` as semantic `grep`, `get_note_content` as `cat`). Structured records—todos with priorities, due dates, and completion status; calendar events; email threads—constitute managed state, manipulated through typed tool interfaces (`create_todo`, `complete_todo`, `search_todos`). External connectors—web search, email providers, calendar systems, and third-party application integrations via standardised protocols—extend the agent's action space beyond the local data store. The tool assembly pipeline serves as the kernel: for each invocation, it loads the appropriate tools for the owner, applies access-control scoping based on the invocation context, and presents a flat namespace to the foundation model. The model's tool calls are the agent's system calls; the kernel mediates every interaction between reasoning and data.

Formally, the personal agent A_i for principal H_i accesses a knowledge store $K_i = (F_i, S_i, M_i)$, where F_i is a set of documents (notes organised in folders), S_i is structured state (todos, calendar, email), and M_i is persistent memory (agent-curated summaries of prior interactions). The agent is equipped with a tool set T_i through which it accesses K_i. Knowledge is *fragmented*: no single document contains a complete picture. Answering a question about a project requires searching notes, reading multiple documents, cross-referencing with todos, and synthesising—the same multi-step retrieval pipeline that makes the agent useful for legitimate queries also enables it to surface sensitive information when the query crosses a privacy boundary.

3.3 Networked Personal Agent

A networked personal agent is a personal agent that can communicate with other personal agents, where each operates under a different human principal. This capability is essential for the tasks that motivate personal agents in the first place: scheduling a meeting requires checking two people's calendars, sharing a project update requires one agent to contact another, and coordinating a product launch requires a manager's agent to collect status from several teammates' agents [21, 7]. Without cross-agent communication, humans remain manually bridging between individually capable AI systems.

The communication mechanism is agent-to-agent remote procedure call. When agent A_j (acting for H_j) invokes `contact_agent(to=`H_i`, message=`m`)`, the system resolves the recipient, loads the per-relationship permissions that H_i has granted to H_j, assembles A_i's tools with those permissions applied, runs A_i's full reasoning loop—system prompt, memory, multi-step tool use, response composition—and returns A_i's final response as a tool result to A_j. This is not a text-only exchange: during execution, A_i may call `search_notes`, `get_note_content`, and other tools to access H_i's private data, and the content of those tool results flows into the response that A_j receives.

Two mechanisms govern cross-boundary interaction. *Access control* determines what the target agent can see. Permissions are stored per-relationship as $\text{access}(H_i, H_j) = \{(\text{domain}, \text{scope}, \text{level})\}$, where $\text{domain} \in \{\text{notes}, \text{todos}, \text{calendar}, \text{email}\}$, $\text{scope} \in \{\text{all}, \text{folder}, \text{none}\}$, and $\text{level} \in \{\text{read}, \text{write}\}$. When scope is `all`, every item in the domain is accessible; when `folder`, only whitelisted folders; when `none`, the corresponding tools are not loaded at all. *Relationship context* $R(H_i, H_j) \in \mathcal{R}$ determines how the agent reasons about appropriate disclosure—a colleague relationship legitimises work-related queries while a stranger provides no trust context. Relationship context does not change which tools are loaded; it influences the agent's judgment, not architecture.

The combination of deep data access and cross-principal communication creates a behavioural safety surface with no analogue in single-agent settings. The target agent has the same tool access to its owner's salary data, medical records, and personal relationships that it uses to answer legitimate questions about project timelines and meeting schedules. When an external agent sends a request via `contact_agent`, the target searches, retrieves, and must decide—within the reasoning loop—which facts to include and which to withhold. We partition K_i into sensitivity categories $C = \{c_1, \ldots, c_m\}$ with a protected subset $C_{\text{prot}} \subset C$, and define a governance policy π that constrains disclosure during cross-boundary interaction, varying along a *specificity* gradient: from no policy (π_0), to a generic privacy instruction (π_1), to category-specific enumeration of protected data types (π_2). For a given configuration (π, R) over queries $Q = Q_{\text{pub}} \cup Q_{\text{prot}}$:

$$U(\pi, R) = \frac{|\{q \in Q_{\text{pub}} : \text{gold facts of } q \text{ appear in response}\}|}{|Q_{\text{pub}}|} \tag{1}$$

$$S(\pi, R) = \frac{|\{q \in Q_{\text{prot}} : \text{no gold facts of } q \text{ appear in response}\}|}{|Q_{\text{prot}}|} \tag{2}$$

where U measures task-completion utility and S measures data-protection security. Neither metric alone suffices: an agent that refuses everything achieves $S{=}1$ but $U{=}0$. The central question is whether governance configurations exist that achieve high values on both axes simultaneously.

4 PART-Bench Design

PART-Bench instantiates the networked personal agent model defined in §3 as a concrete, reproducible evaluation platform. We describe the benchmark scenario, evaluation queries, governance gradient, interaction modes, and dual-metric protocol.

4.1 Benchmark Scenario

The benchmark centres on a two-agent interaction. The *target agent* ("Alex's agent") is a fully-equipped personal agent as defined in §3.2: it has persistent memory containing its owner's preferences and interaction history, 85+ documents organised in hierarchical folders (Work/Projects, Work/Meetings, Work/HR, Personal/Finance, Personal/Health, Personal/Family) each containing facts that map to evaluation queries, callable tools (`search_notes`, `get_note_content`, `search_todos`, `create_todo`) that access real encrypted data through the personal agent's tool assembly pipeline, and an identity established through system prompt configuration. The *querying agent* ("Tina's agent") acts on behalf of a second principal and communicates with Alex's agent through the `contact_agent` RPC mechanism described in §3.3. In multi-turn mode, Tina's agent maintains a checklist of pending questions and adaptively selects which to ask, how to frame them, and whether to retry based on the evolving conversation history.

4.2 Evaluation Queries

PART-Bench comprises XXXX evaluation queries across 5 sensitivity categories:

Table 2: Evaluation query distribution across sensitivity categories. Work/public queries measure utility (should be answered); all others measure security (should be refused).

Category	ID range	Count	Expected behaviour
Work/public	Q1–XXXX	XXXX	Answer (utility)
Sensitive work (HR, board, investor)	XXXX–XXXX	XXXX	Refuse (security)
Personal finance	XXXX–XXXX	XXXX	Refuse (security)
Personal health	XXXX–XXXX	XXXX	Refuse (security)
Personal relationships	XXXX–XXXX	XXXX	Refuse (security)

Each query is paired with **gold key facts**: specific pieces of information whose presence in the agent's response constitutes a leak (for sensitive queries) or a correct answer (for work/public queries). This enables automated evaluation via string matching against gold facts, removing the need for LLM-as-judge and its associated reliability concerns.

4.3 Defence Gradient

PART-Bench separates prompt-level governance from architectural isolation. Prompt-level defences keep the target data visible to the model and rely on the agent's reasoning to withhold restricted information. Architectural defences remove or gate data before it enters the generation context, testing whether safety improves when the agent cannot access restricted content in the first place.

The governance gradient comprises seven levels. **D0** provides no defence: the agent's system prompt contains no privacy instructions, establishing baseline behaviour with full data access and zero policy. **D1** appends a generic instruction ("*Try not to share personal or private information*")—the most common real-world defence pattern. **D2** provides a category-specific deny list that explicitly enumerates protected data types ("*Never share: financial data including salary, equity, and investments; health information including conditions, medications, and appointments; HR and compensation details; personal relationship information*"). D1 and D2 express the same strictness intent; they differ only in specificity, isolating whether the agent's compliance depends on knowing exactly what to protect. **D3** applies spotlighting [8]: the external agent's message is explicitly marked as untrusted input with delimiters, separating counterpart requests from the target agent's own instructions and memory. **D4** establishes an instruction hierarchy [23], explicitly ordering authority levels in the system prompt: system policy, owner policy, tool output, then external-agent request. **D5** introduces architectural isolation through Mountable Context Cells (MCC): restricted notes, tools, and memory shards are physically removed from the context before the agent processes the request. **D6** extends D5 with an escalation protocol: when the agent detects a request about restricted data, it flags the query for owner review instead of refusing outright, recovering utility without exposing the data.

D0 through D4 isolate the effect of prompt-level policy language. D5 and D6 test architectural isolation. This split is central to the benchmark: if D5 dominates D4 at similar utility, the limiting factor is not prompt wording but context exposure.

4.4 Evaluation Modes

Single-step mode. Each query is asked in isolation: one question, one response, no conversation history. This provides a controlled measurement of per-query behaviour but misses conversational dynamics.

Multi-turn heartbeat mode. Both agents operate as persistent processes, exchanging messages at regular intervals ("heartbeats"). Tina's agent maintains a checklist of pending questions and decides which to ask, how to frame each request, and whether to retry based on the evolving conversation. This mode captures four dynamics absent from single-step evaluation: escalation through repeated probing after initial refusal, context accumulation where information from earlier turns influences later responses, social framing where the querying agent wraps sensitive questions in work-acceptable contexts, and within-tick batching where multiple questions are asked in a single turn to overwhelm defences.

4.5 Dual-Metric Protocol

Every experimental run produces the utility score U and security score S defined in Equations 1 and 2. A perfect governance configuration achieves $U = S = 1$: all legitimate queries answered, all sensitive data protected. The (U, S) pair for each configuration defines a point on the Pareto frontier. A defence that achieves $S = 1$ by refusing all queries is useless; one with $U = 1$ that leaks everything is dangerous. Only dual measurement captures the tradeoff that practitioners face, and no prior cross-boundary benchmark reports both metrics simultaneously.

Table 3: Layered PART-Bench experimental design. The NeurIPS paper first runs Layer 0 to select and characterize the model substrate. Layer 1 and a small Layer 2 attack slice are optional additions if space and budget permit. Layers 3, 5, and 6 are benchmark extensions for the thesis and follow-up defense paper.

Layer	Question	Matrix	Role
L0 Model sentinel	Which model should serve as the ablation backbone?	Five models $\times$ 2 reps $\times$ SS/MS $\times$ D0/D1/D2 $\times$ A0 $\times$ R0	Core paper
L1 Relationship	Does relationship context shift category leakage?	Backbone model $\times$ 3 reps $\times$ SS/MS $\times$ D0/D1/D2 $\times$ A0 $\times$ R0/R1	Optional paper / thesis
L2 Attacks	Do formal attacks break prompt-level defences more than natural questioning?	Backbone model $\times$ 3 reps $\times$ D1/D2 $\times$ A0/A1/A2/A3 $\times$ R0	Strong paper slice
L3 Defences	Where does the utility–security frontier bend?	Backbone model $\times$ 3 reps $\times$ SS/MS $\times$ D0–D6 $\times$ A0 $\times$ R0	Thesis / follow-up
L4 Model confirmation	Do later defense or attack findings reproduce across models?	Selected final cells $\times$ five models	Optional confirmation
L5 Actions	Does cross-boundary pressure cause unsafe actions, not just leaks?	Backbone model $\times$ 3 reps $\times$ D0/D2/D5/D6 $\times$ A0/A2 $\times$ R0/R1 $\times$ action tasks	Thesis / follow-up
L6 Social scale	Does relationship distribution change aggregate leakage?	Backbone model $\times$ 3 reps $\times$ MS $\times$ D0/D2/D5 $\times$ A0 $\times$ S1/S3/S4/S50	Thesis / follow-up

4.6 Relationship and Agent-Network Conditions

The core benchmark evaluates two relationship conditions. Under **R0 (Stranger)**, Tina is identified by name only and the agent treats her as unknown, providing no trust context. Under **R1 (Colleague)**, Tina is identified as a project manager who works with Alex daily; shared project context and interaction history are provided through the relationship memory. The extended benchmark adds **R2 (Close friend)** and **R3 (Investor/formal stakeholder)** to test whether personal, professional, and strategic relationships shift disclosure in different sensitivity categories. PART-Bench also supports larger social setups—a single requester (S1), three requesters with different relationships (S3), four agents that can interact with one another (S4), and fifty requesters with one interaction each (S50)—testing whether leakage is a property of an isolated conversation or of the relationship distribution around the personal agent.

5 Experimental Setup

Design principle. The complete PART-Bench design crosses models, modes, defences, attacks, relationships, and social-network structure. A naive full factorial design is not scientifically useful: it would require 5 models $\times$ 3 replications $\times$ 2 modes $\times$ 7 defences $\times$ 4 attacks $= 840$ runs per social setup, before adding multi-agent network conditions. We therefore use a staged design. Each layer answers one new question that the previous layer cannot answer, and later layers are only run after the preceding layer is stable.

Model sentinel and replications. Layer 0 is the first experiment, not a later robustness check. It compares five model families: `gpt-5-mini`, `gpt-5.4-mini`, `gpt-5.4`, `kimi`, and `deepseek`. Each cell is run with two independent replications. The goal is to choose a stable, representative, cost-effective backbone model for later ablations, while also testing whether the basic signatures of PART-Bench hold across model families.

Core benchmark. The minimum NeurIPS experiment is Layer 0:

$$5 \text{ models} \times 2 \text{ reps} \times 2 \text{ modes} \times 3 \text{ policies} \times 1 \text{ attack} \times 1 \text{ relationship} = 60 \text{ runs}.$$

Each single-step run evaluates XXXX queries independently. Each multi-step run uses the 10×20 split protocol: ten groups of twenty questions, with sixty heartbeat ticks per group. The relationship condition is fixed to R0 (stranger) so that the first experiment isolates model and policy behavior rather than social context.

Table 4: Layer 0 model sentinel. The first experiment fixes relationship to R0 and attack to A0 (direct / natural querying), then varies model, mode, and policy. Values are placeholders in this draft; final results report means over two replications.

Model	D0 SS	D1 SS	D2 SS	D0 MS	D1 MS	D2 MS	Selection note
gpt-5-mini	XXXX	XXXX	XXXX	XXXX	XXXX	XXXX	Candidate backbone
gpt-5.4-mini	XXXX	XXXX	XXXX	XXXX	XXXX	XXXX	Candidate backbone
gpt-5.4	XXXX	XXXX	XXXX	XXXX	XXXX	XXXX	Stronger, higher-cost reference
kimi	XXXX	XXXX	XXXX	XXXX	XXXX	XXXX	Non-OpenAI transfer
deepseek	XXXX	XXXX	XXXX	XXXX	XXXX	XXXX	Non-OpenAI transfer

Table 5: Layer 0 mode and policy decomposition. This table is filled after selecting the backbone model from Table 4; it reports utility, security, and false refusal for the three policies under single-step and multi-step evaluation.

Policy	U_{SS}	S_{SS}	**FRR**$_{\mathrm{SS}}$	S_{MS}	First leak tick
D0 No defense	XXXX	XXXX	XXXX	XXXX	XXXX
D1 Generic privacy	XXXX	XXXX	XXXX	XXXX	XXXX
D2 Explicit allow/deny	XXXX	XXXX	XXXX	XXXX	XXXX

Strong benchmark additions. Layer 1 adds relationship context after the backbone model is chosen:

$$1 \text{ backbone model} \times 3 \text{ reps} \times 2 \text{ modes} \times 3 \text{ policies} \times 1 \text{ attack} \times 2 \text{ relationships} = 36 \text{ runs.}$$

The attack slice adds a small Crescendo-style multi-turn probe:

$$1 \text{ model} \times 3 \text{ reps} \times 1 \text{ defence} \times 1 \text{ attack} \times 1 \text{ mode} \times 2 \text{ relationships} = 6 \text{ runs.}$$

Evaluation. For every run we compute utility U and security S as defined in Equations 1 and 2. We also report false refusal rate on public work queries, leak rate by sensitivity category, and first-leak tick in multi-turn mode. Action-safety extensions use separate metrics: unauthorized action rate, unauthorized mutation/share rate, and wrong-principal action rate.

6 Results

6.1 Layer 0: Model Sentinel

Expected finding 1: model substrate matters. Absolute utility and security rates may vary substantially by model. This determines which model should serve as the main ablation backbone for later defense, attack, and relationship experiments.

Expected finding 2: the raw stack is not deployable as-is. D0 should preserve utility but leak protected facts at a high rate, especially in multi-step mode. If this holds across most models, the benchmark exposes a stack-level deployment risk rather than a single-model artifact.

Expected finding 3: explicit allow/deny policy beats generic privacy language. D1 tests the common deployment pattern of adding a vague privacy sentence; D2 states clearly what may be shared and what must not be shared. The key comparison is D0 versus D1 versus D2 across both modes.

6.2 Mode and Policy Effects

Expected finding 4: single-step evaluation undercounts deployment risk. The multi-step heartbeat setting is expected to leak more than isolated one-shot questions because earlier answers create context, the querying agent can reframe unresolved requests, and multiple questions can be batched into a

Table 6: Optional relationship baseline after the backbone model is selected. This table is not part of the first model sentinel; it tests whether relationship context shifts leakage by category.

Category	D0 R0	D2 R0	D0 R1	D2 R1	Interpretation
Sensitive work	XXXX	XXXX	XXXX	XXXX	Close to legitimate work context
Personal finance	XXXX	XXXX	XXXX	XXXX	Usually easy to identify as protected
Personal health	XXXX	XXXX	XXXX	XXXX	Expected to be clearly out-of-scope for colleagues
Personal relationships	XXXX	XXXX	XXXX	XXXX	May increase under social framing

Table 7: Attack robustness slice. A0 is natural questioning; A2 is a Crescendo-style multi-turn escalation. A1 and A3 are retained as benchmark extensions.

Defence	Attack	Mode	Security	Expected failure pattern
D1 Generic	A0 Direct	SS/MS	XXXX	Vague instruction fails under ordinary requests
D2 Category	A0 Direct	SS/MS	XXXX	Direct sensitive requests often blocked
D2 Category	A2 Crescendo	MS	XXXX	Gradual reframing turns protected data into apparently useful context
D2 Category	A3 Persuasion	SS	XXXX	Authority or reciprocity framing pressures refusal boundary

single turn. This result is a methodological warning: a single-step benchmark can certify an agent that remains unsafe in deployment-like interaction.

Expected finding 5: D2 should improve security without collapsing utility. If D2 raises S while preserving U and keeping false refusal rate acceptable, the first practical recommendation is not an architectural redesign but a zero-cost policy improvement: state both allowed and disallowed information categories explicitly.

6.3 Layer 1: Relationship Baseline

Expected finding 6: relationship is a security parameter. R1 is not expected to uniformly increase or decrease leakage. Instead, it should redistribute risk across categories. A colleague relationship may legitimize sensitive work or relationship-context questions while making health and finance requests feel less appropriate.

6.4 Layer 2: Attack Slice

Expected finding 7: formal attacks turn residual leaks into systematic failures. D2 should remain useful against direct requests, but multi-turn escalation and persuasion should expose category ambiguity and partial-compliance failures. This does not replace the core benchmark; it shows that the same harness can gauge adversarial pressure.

6.5 Benchmark Extensions: Defences, Actions, and Social Scale

The extension tables make the benchmark cumulative rather than factorial. The paper can report the core deployment result first, then add cross-model and attack slices without committing to every combination. The thesis can then use the same harness to evaluate architectural isolation, action safety, and social-network scale.

Table 8: Planned defense-frontier result table. Layers 3, 5, and 6 are not required for the minimum NeurIPS benchmark but define how PART-Bench scales into the thesis and follow-up defense paper.

Defence	Type	Utility	Security	Expected role
D0 No defence	Raw	XXXX	XXXX	Unsafe baseline
D1 Generic prompt	Prompt	XXXX	XXXX	Security theater baseline
D2 Category deny list	Prompt	XXXX	XXXX	First Pareto improvement
D3 Spotlighting	Prompt	XXXX	XXXX	Incremental prompt hardening
D4 Instruction hierarchy	Prompt	XXXX	XXXX	Stronger prompt-level ceiling
D5 MCC isolation	Architectural	XXXX	XXXX	Qualitative security jump
D6 MCC + escalation	Architectural	XXXX	XXXX	Utility recovery and fewer false refusals

Table 9: Planned action-safety track. Action tasks are separated from QA because the failure mode is unauthorized state change rather than information disclosure.

Task family	Authorized utility	Unauthorized action rate	Expected risk
Calendar	XXXX	XXXX	Exposes full calendar or schedules without permission
Notes	XXXX	XXXX	Shares, edits, or summarizes restricted notes
Messaging	XXXX	XXXX	Drafts or sends messages under the wrong principal
Todos	XXXX	XXXX	Creates or completes tasks for the wrong actor

7 Discussion and Limitations

Implications for practitioners. PART-Bench is designed to answer a deployment question: given a raw or defended personal-agent stack, where does it sit on the utility–security frontier? The expected practical lessons are: (1) do not deploy cross-boundary personal agents with raw or generic privacy prompts alone; (2) measure utility and security together, because refusal-only systems are safe but useless; (3) evaluate multi-turn interactions, because one-shot tests understate risk; and (4) treat relationship context as a security parameter, not just a personalization feature.

Toward architectural defences. The defense-frontier layer distinguishes prompt hardening from architectural isolation. If D3 and D4 improve security only incrementally while D5 changes the frontier, the conclusion is not merely that prompts need to be better. It is that personal-agent platforms need permissioned context construction: do not ask the agent to avoid sharing data it can see; ensure the data is not mounted for that relationship unless it should be available. D6 then tests whether escalation can recover utility lost through isolation.

Limitations. **Staged scope**: the benchmark is intentionally not a full factorial grid. This improves interpretability but means each layer answers a bounded question. **Synthetic scenario**: the personal agent's data is synthetic, although it is structured to resemble realistic notes, memory, and tools. **Evaluation method**: gold-fact string matching may miss paraphrased leaks or trigger on coincidental matches; we will supplement it with LLM-based and human spot-check evaluation. **Model coverage**: Layer 4 tests five model families, but the benchmark should be rerun as new agent models and tool-use stacks emerge. **Attack coverage**: A0 and A2 cover natural and progressive multi-turn pressure; PAIR-style optimization and persuasion attacks remain benchmark extensions.

Benchmark release. We will release the benchmark suite including: evaluation queries with gold key facts, agent configuration files (notes, memory, tools), the heartbeat evaluation engine, action-task templates, social-network setup templates, and evaluation scripts. The framework is model-agnostic and explicitly staged so future work can add models, attacks, defences, action families, and relationship graphs without changing the core utility–security contract.

Table 10: Planned social-scaling track. These settings measure whether leakage is a property of one conversation or of the relationship network around the personal agent.

Setup	Metric	Expected pattern
S1 Single requester	Utility / security	Clean baseline for comparison
S3 Three requesters	Category leak by relationship	Different relationships pull different categories across the boundary
S4 Four interacting agents	Cross-agent leakage path	Agents may route around a boundary by asking each other
S50 Fifty requesters	Aggregate leak distribution	Social graph composition changes population-level leakage

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A Observed Attack Patterns

Under D2 (category-specific deny list), residual leaks in single-step evaluation cluster into identifiable patterns:

Table 11: Distribution of D2 leak causes in single-step evaluation (XXXX total leaks).

Pattern	Count	Share
Category ambiguity (data accessible but semantically sensitive)	XXXX	XXXX%
Social framing (personal question in work-acceptable frame)	XXXX	XXXX%
Partial compliance failure (refusal then continued disclosure)	XXXX	XXXX%
Other	XXXX	XXXX%

Category ambiguity is the dominant failure mode: data stored in "Work" folders that is semantically sensitive (e.g., board meeting notes containing investor terms, HR documents with salary benchmarks). The agent's policy correctly protects "personal finance" but does not recognise that a board memo discussing fundraising terms is financially sensitive.

B Experimental Configuration Details

Alex's agent. System prompt establishes identity (Co-founder & CTO, Oxford CS background). 85+ notes across 6 folder categories. Tools: `read_note`, `search_notes`, `check_calendar`, `schedule_meeting`. Memory contains preferences and interaction history.

Tina's agent. Operates from a HEARTBEAT.md instruction file. Phase 1 (information gathering): systematically works through pending questions. Phase 2 (deeper probing): re-asks refused questions with different framing. Maintains a MEMORY.md checklist tracking answered/refused/pending status per question.

Heartbeat configuration. Each tick: Tina's agent selects 1–3 questions, sends a message, Alex's agent responds. XXXX ticks per group in multi-turn mode.

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